

Administrator Operations Manual

Comprehensive Guide to User Management, Security Controls, openpyxl Excel Exports, and Database Recovery

Document Title:	Administrator Operations Manual
Target Audience:	System Administrators, DevOps Engineers, Security Officers
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Access Requirement:	OAuth2 Bearer JWT Token with Elevated `Admin` Role Scope

RESTRICTED ACCESS NOTICE: The procedures within this manual govern elevated system administrative rights and database recovery operations. Distribution is strictly restricted to authorized system administrators.

1.0 DOCUMENT REVISION HISTORY

Rev #	Date	Author	Reviewer	Approver	Description of Changes
1.0	2026-08-01	Ashwindh	Narendar R.	Infosys Mentor	Initial System Administrator Manual Draft
1.5	2026-08-04	Ashwindh	Pravallika	Technical Lead	Added openpyxl Export & Database Backup Protocol
2.0	2026-08-07	Team PricePilot	QA Team	Academic Committee	Final Administrator Operations Manual Validation

2.0 ADMINISTRATIVE GOVERNANCE & RBAC

PricePilot AI enforces strict Role-Based Access Control (RBAC). Administrative privileges are granted exclusively to user accounts where ``role == 'Admin'``. Unauthenticated or standard merchant users (``role == 'User'``) attempting to access administrative endpoints receive an immediate ``403 Forbidden`` response.

System Capability / Route	Standard User Scope	Admin Scope (<code>`role == 'Admin'`</code>)
AI Price Predictions (<code>`/predict`</code>)	Permitted	Permitted
Prediction History (<code>`/history`</code>)	Own Predictions Only	System-Wide Predictions
User Management Dashboard (<code>`/users`</code>)	Denied (403)	Full Control (Create, Update, Delete)
Bulk User Operations (<code>`bulk-status`</code>)	Denied (403)	Full Control
openpyxl Excel Export (<code>`export-users`</code>)	Denied (403)	Full Control
Database Backup & System Config	Denied (403)	Full Control

3.0 USER ADMINISTRATION & BULK OPERATIONS

Administrators manage user account lifecycles via ``/users``:

- 1. **Account Approval:** Change newly registered user status from ``pending`` to ``approved``.
- 2. **Account Blocking:** Set status to ``blocked`` to immediately revoke access.
- 3. **Bulk Actions:** Select multiple user rows and trigger ``POST /api/users/bulk-status`` or ``POST /api/users/bulk-delete``.

4.0 OPENPYXL EXCEL DATA EXPORTER ENGINE

Administrators can generate native, styled Excel workbooks (`Users\_Report.xlsx`) by calling `GET /api/admin/export-users`.

Excel Styling & Formatting Specifications:

- **Header Style:** Dark Blue Fill (`#1E3A8A`), Bold White Text (Helvetica 11pt).
- **Data Rows:** Alternating Zebra Striping (`#F8FAFC`), Thin Gray Borders (`#CBD5E1`).
- **Layout Rules:** Freeze Panes on Header Row (A4), Auto-Filters on all columns, Auto-Adjusted Column Widths (+4 padding).

5.0 DATABASE BACKUP & DISASTER RECOVERY RUNBOOK

To perform an immediate PostgreSQL database backup:

```
# PostgreSQL Neon Cloud Backup Command pg_dump $DATABASE_URL --format=custom --file=pricepilot_backup_$(date +%Y%m%d).dump # SQLite Local Backup Fallback cp backend/pricepilot.db backend/backups/pricepilot_$(date +%Y%m%d).db
```

To restore a database dump:

```
# Restore PostgreSQL Dump Command pg_restore --clean --dbname=$DATABASE_URL pricepilot_backup_20260807.dump
```

6.0 ADMIN TROUBLESHOOTING & AUDIT LOGGING

Audit logs are stored in the `ActivityLog` table tracking user ID, action string, IP address, and timestamp. System Administrators should audit logs weekly for suspicious brute-force login attempts.

Incident Condition	Root Cause Analysis	System Admin Resolution Action
403 Forbidden on Export	JWT token lacks Admin role	Promote user role to 'Admin' in `users` database table.
Database Lock Error	Concurrent SQLite write locks	Migrate environment variable `DATABASE_URL` to Neon PostgreSQL.
ML Prediction Timeout	Memory allocation shortage	Ensure server has at least 2GB RAM for `best_price_model.pkl`.